

CHAPTER 13—ANALYSIS OF VARIANCE AND EXPERIMENTAL DESIGN

MULTIPLE CHOICE

1. In the analysis of variance procedure (ANOVA), "*factor*" refers to
 - a. the dependent variable
 - b. the independent variable
 - c. different levels of a treatment
 - d. the critical value of F
2. The ANOVA procedure is a statistical approach for determining whether or not
 - a. the means of two samples are equal
 - b. the means of two or more samples are equal
 - c. the means of more than two samples are equal
 - d. the means of two or more populations are equal
3. The variable of interest in an ANOVA procedure is called
 - a. a partition
 - b. a treatment
 - c. either a partition or a treatment
 - d. a factor
4. In the ANOVA, treatment refers to
 - a. experimental units
 - b. different levels of a factor
 - c. the dependent variable
 - d. applying antibiotic to a wound
5. In factorial designs, the response produced when the treatments of one factor interact with the treatments of another in influencing the response variable is known as
 - a. main effect
 - b. replication
 - c. interaction
 - d. None of these alternatives is correct.
6. An experimental design where the experimental units are randomly assigned to the treatments is known as
 - a. factor block design
 - b. random factor design
 - c. completely randomized design
 - d. None of these alternatives is correct.
7. The number of times each experimental condition is observed in a factorial design is known as
 - a. partition
 - b. replication
 - c. experimental condition
 - d. factor
8. In an analysis of variance problem involving 3 treatments and 10 observations per treatment, $SSE = 399.6$. The MSE for this situation is
 - a. 133.2
 - b. 13.32
 - c. 14.8
 - d. 30.0

9. When an analysis of variance is performed on samples drawn from K populations, the mean square between treatments (MSTR) is
 - a. $SSTR/n_T$
 - b. $SSTR/(n_T - 1)$
 - c. $SSTR/K$
 - d. $SSTR/(K - 1)$
10. In an analysis of variance where the total sample size for the experiment is n_T and the number of populations is K , the mean square within treatments is
 - a. $SSE/(n_T - K)$
 - b. $SSTR/(n_T - K)$
 - c. $SSE/(K - 1)$
 - d. SSE/K
11. The F ratio in a completely randomized ANOVA is the ratio of
 - a. $MSTR/MSE$
 - b. MST/MSE
 - c. $MSE/MSTR$
 - d. MSE/MST
12. The critical F value with 6 numerator and 60 denominator degrees of freedom at $\alpha = .05$ is
 - a. 3.74
 - b. 2.25
 - c. 2.37
 - d. 1.96
13. An ANOVA procedure is applied to data obtained from 6 samples where each sample contains 20 observations. The degrees of freedom for the critical value of F are
 - a. 6 numerator and 20 denominator degrees of freedom
 - b. 5 numerator and 20 denominator degrees of freedom
 - c. 5 numerator and 114 denominator degrees of freedom
 - d. 6 numerator and 20 denominator degrees of freedom
14. The mean square is the sum of squares divided by
 - a. the total number of observations
 - b. its corresponding degrees of freedom
 - c. its corresponding degrees of freedom minus one
 - d. None of these alternatives is correct.
15. In an analysis of variance problem if $SST = 120$ and $SSTR = 80$, then SSE is
 - a. 200
 - b. 40
 - c. 80
 - d. 120
16. The required condition for using an ANOVA procedure on data from several populations is that the
 - a. the selected samples are dependent on each other
 - b. sampled populations are all uniform
 - c. sampled populations have equal variances
 - d. sampled populations have equal means
17. An ANOVA procedure is used for data that was obtained from four sample groups each comprised of five observations. The degrees of freedom for the critical value of F are
 - a. 3 and 20
 - b. 3 and 16

- c. 4 and 17
 - d. 3 and 19
18. In ANOVA, which of the following is not affected by whether or not the population means are equal?
- a. $\bar{x}$
 - b. between-samples estimate of σ^2
 - c. within-samples estimate of σ^2
 - d. None of these alternatives is correct.
19. A term that means the same as the term "variable" in an ANOVA procedure is
- a. factor
 - b. treatment
 - c. replication
 - d. variance within
20. In order to determine whether or not the means of two populations are equal,
- a. a t test must be performed
 - b. an analysis of variance must be performed
 - c. either a t test or an analysis of variance can be performed
 - d. a chi-square test must be performed
21. The process of allocating the total sum of squares and degrees of freedom is called
- a. factoring
 - b. blocking
 - c. replicating
 - d. partitioning
22. An experimental design that permits statistical conclusions about two or more factors is a
- a. randomized block design
 - b. factorial design
 - c. completely randomized design
 - d. randomized design
23. In a completely randomized design involving three treatments, the following information is provided:

	Treatment 1	Treatment 2	Treatment 3
Sample Size	5	10	5
Sample Mean	4	8	9

The overall mean for all the treatments is

- a. 7.00
 - b. 6.67
 - c. 7.25
 - d. 4.89
24. In a completely randomized design involving four treatments, the following information is provided.

	Treatment 1	Treatment 2	Treatment 3	Treatment 4
Sample Size	50	18	15	17
Sample Mean	32	38	42	48

The overall mean (the grand mean) for all treatments is

- a. 40.0
- b. 37.3
- c. 48.0

- d. 37.0
25. An ANOVA procedure is used for data obtained from five populations. five samples, each comprised of 20 observations, were taken from the five populations. The numerator and denominator (respectively) degrees of freedom for the critical value of F are
- 5 and 20
 - 4 and 20
 - 4 and 99
 - 4 and 95
26. The critical F value with 8 numerator and 29 denominator degrees of freedom at $\alpha = 0.01$ is
- 2.28
 - 3.20
 - 3.33
 - 3.64
27. An ANOVA procedure is used for data obtained from four populations. Four samples, each comprised of 30 observations, were taken from the four populations. The numerator and denominator (respectively) degrees of freedom for the critical value of F are
- 3 and 30
 - 4 and 30
 - 3 and 119
 - 3 and 116
28. Which of the following is **not** a required assumption for the analysis of variance?
- The random variable of interest for each population has a normal probability distribution.
 - The variance associated with the random variable must be the same for each population.
 - At least 2 populations are under consideration.
 - Populations have equal means.
29. In an analysis of variance, one estimate of σ^2 is based upon the differences **between** the treatment means and the
- means of each sample
 - overall sample mean
 - sum of observations
 - populations have equal means

NARRBEGIN: Exhibit 13-1

Exhibit 13-1

$$SSTR = 6,750$$

$$SSE = 8,000$$

$$n_T = 20$$

$$H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4$$

$$H_a: \text{at least one mean is different}$$

NARREND

30. Refer to Exhibit 13-1. The mean square between treatments (MSTR) equals
- 400
 - 500
 - 1,687.5
 - 2,250
31. Refer to Exhibit 13-1. The mean square within treatments (MSE) equals
- 400

- b. 500
 - c. 1,687.5
 - d. 2,250
32. Refer to Exhibit 13-1. The test statistic to test the null hypothesis equals
- a. 0.22
 - b. 0.84
 - c. 4.22
 - d. 4.5
33. Refer to Exhibit 13-1. The null hypothesis is to be tested at the 5% level of significance. The p -value is
- a. less than .01
 - b. between .01 and .025
 - c. between .025 and .05
 - d. between .05 and .10
34. Refer to Exhibit 13-1. The null hypothesis
- a. should be rejected
 - b. should not be rejected
 - c. was designed incorrectly
 - d. None of these alternatives is correct.

NARRBEGIN: Exhibit 13-2

Exhibit 13-2

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F
Between Treatments	2,073.6	4		
Between Blocks	6,000	5	1,200	
Error		20	288	
Total		29		

NARREND

35. Refer to Exhibit 13-2. The null hypothesis for this ANOVA problem is
- a. $\mu_1 = \mu_2 = \mu_3 = \mu_4$
 - b. $\mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5$
 - c. $\mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5 = \mu_6$
 - d. $\mu_1 = \mu_2 = \dots = \mu_{20}$
36. Refer to Exhibit 13-2. The mean square between treatments equals
- a. 288
 - b. 518.4
 - c. 1,200
 - d. 8,294.4
37. Refer to Exhibit 13-2. The sum of squares due to error equals
- a. 14.4
 - b. 2,073.6
 - c. 5,760
 - d. 6,000

38. Refer to Exhibit 13-2. The test statistic to test the null hypothesis equals
 - a. 0.432
 - b. 1.8
 - c. 4.17
 - d. 28.8
39. Refer to Exhibit 13-2. The null hypothesis is to be tested at the 5% level of significance. The p -value is
 - a. greater than 0.10
 - b. between 0.10 to 0.05
 - c. between 0.05 to 0.025
 - d. between 0.025 to 0.01
40. Refer to Exhibit 13-2. The null hypothesis
 - a. should be rejected
 - b. should not be rejected
 - c. should be revised
 - d. None of these alternatives is correct.

NARRBEGIN: Exhibit 13-3

Exhibit 13-3

To test whether or not there is a difference between treatments A, B, and C, a sample of 12 observations has been randomly assigned to the 3 treatments. You are given the results below.

Treatment		Observation		
A	20	30	25	33
B	22	26	20	28
C	40	30	28	22

NARREND

41. Refer to Exhibit 13-3. The null hypothesis for this ANOVA problem is
 - a. $\mu_1 = \mu_2$
 - b. $\mu_1 = \mu_2 = \mu_3$
 - c. $\mu_1 = \mu_2 = \mu_3 = \mu_4$
 - d. $\mu_1 = \mu_2 = \dots = \mu_{12}$
42. Refer to Exhibit 13-3. The mean square between treatments (MSTR) equals
 - a. 1.872
 - b. 5.86
 - c. 34
 - d. 36
43. Refer to Exhibit 13-3. The mean square within treatments (MSE) equals
 - a. 1.872
 - b. 5.86
 - c. 34
 - d. 36
44. Refer to Exhibit 13-3. The test statistic to test the null hypothesis equals
 - a. 0.944
 - b. 1.059
 - c. 3.13

- d. 19.231
45. Refer to Exhibit 13-3. The null hypothesis is to be tested at the 1% level of significance. The p -value is
- greater than 0.1
 - between 0.1 to 0.05
 - between 0.05 to 0.025
 - between 0.025 to 0.01
46. Refer to Exhibit 13-3. The null hypothesis
- should be rejected
 - should not be rejected
 - should be revised
 - None of these alternatives is correct.

NARRBEGIN: Exhibit 13-4

Exhibit 13-4

In a completely randomized experimental design involving five treatments, 13 observations were recorded for each of the five treatments (a total of 65 observations). The following information is provided.

$SSTR = 200$ (Sum Square Between Treatments)

$SST = 800$ (Total Sum Square)

NARREND

47. Refer to Exhibit 13-4. The sum of squares within treatments (SSE) is
- 1,000
 - 600
 - 200
 - 1,600
48. Refer to Exhibit 13-4. The number of degrees of freedom corresponding to between treatments is
- 60
 - 59
 - 5
 - 4
49. Refer to Exhibit 13-4. The number of degrees of freedom corresponding to within treatments is
- 60
 - 59
 - 5
 - 4
50. Refer to Exhibit 13-4. The mean square between treatments (MSTR) is
- 3.34
 - 10.00
 - 50.00
 - 12.00
51. Refer to Exhibit 13-4. The mean square within treatments (MSE) is
- 50
 - 10

- c. 200
 - d. 600
52. Refer to Exhibit 13-4. The test statistic is
- a. 0.2
 - b. 5.0
 - c. 3.75
 - d. 15
53. Refer to Exhibit 13-4. If at 95% confidence we want to determine whether or not the means of the five populations are equal, the p -value is
- a. between 0.05 to 0.10
 - b. between 0.025 to 0.05
 - c. between 0.01 to 0.025
 - d. less than 0.01

NARRBEGIN: Exhibit 13-5

Exhibit 13-5

Part of an ANOVA table is shown below.

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F
Between Treatments	180	3		
Within Treatments (Error)				
TOTAL	480	18		

NARREND

54. Refer to Exhibit 13-5. The mean square between treatments (MSTR) is
- a. 20
 - b. 60
 - c. 300
 - d. 15
55. Refer to Exhibit 13-5. The mean square within treatments (MSE) is
- a. 60
 - b. 15
 - c. 300
 - d. 20
56. Refer to Exhibit 13-5. The test statistic is
- a. 2.25
 - b. 6
 - c. 2.67
 - d. 3
57. Refer to Exhibit 13-5. If at 95% confidence, we want to determine whether or not the means of the populations are equal, the p -value is
- a. between 0.01 to 0.025

- b. between 0.025 to 0.05
- c. between 0.05 to 0.1
- d. greater than 0.1

NARRBEGIN: Exhibit 13-6

Exhibit 13-6

Part of an ANOVA table is shown below.

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F
Between Treatments	64			8
Within Treatments			2	
Error				
Total	100			

NARREND

58. Refer to Exhibit 13-6. The number of degrees of freedom corresponding to between treatments is
 - a. 18
 - b. 2
 - c. 4
 - d. 3
59. Refer to Exhibit 13-6. The number of degrees of freedom corresponding to within treatments is
 - a. 22
 - b. 4
 - c. 5
 - d. 18
60. Refer to Exhibit 13-6. The mean square between treatments (MSTR) is
 - a. 36
 - b. 16
 - c. 64
 - d. 15
61. Refer to Exhibit 13-6. If at 95% confidence we want to determine whether or not the means of the populations are equal, the p -value is
 - a. greater than 0.1
 - b. between 0.05 to 0.1
 - c. between 0.025 to 0.05
 - d. less than 0.01
62. Refer to Exhibit 13-6. The conclusion of the test is that the means
 - a. are equal
 - b. may be equal
 - c. are not equal
 - d. None of these alternatives is correct.

NARRBEGIN: Exhibit 13-7

Exhibit 13-7

The following is part of an ANOVA table that was obtained from data regarding three treatments and a total of 15 observations.

Source of Variation	Sum of Squares	Degrees of Freedom
Between Treatments	64	
Error (Within Treatments)	96	

NARREND

63. Refer to Exhibit 13-7. The number of degrees of freedom corresponding to between treatments is
 - a. 12
 - b. 2
 - c. 3
 - d. 4
64. Refer to Exhibit 13-7. The number of degrees of freedom corresponding to within treatments is
 - a. 12
 - b. 2
 - c. 3
 - d. 15
65. Refer to Exhibit 13-7. The mean square between treatments (MSTR) is
 - a. 36
 - b. 16
 - c. 8
 - d. 32
66. Refer to Exhibit 13-7. The computed test statistics is
 - a. 32
 - b. 8
 - c. 0.667
 - d. 4
67. Refer to Exhibit 13-7. If at 95% confidence, we want to determine whether or not the means of the populations are equal, the p -value is
 - a. between 0.01 to 0.025
 - b. between 0.025 to 0.05
 - c. between 0.05 to 0.1
 - d. greater than 0.1
68. Refer to Exhibit 13-7. The conclusion of the test is that the means
 - a. are equal
 - b. may be equal
 - c. are not equal
 - d. None of these alternatives is correct.

PROBLEM

1. Information regarding the ACT scores of samples of students in three different majors is given below.

Major

	Management	Finance	Accounting
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Sample size	12	9	12
Average	24	25	26
Sample variance	18	7	10

- a. Compute the overall sample mean $\bar{X}$.
 - b. Set up the ANOVA table for this problem including the test statistic.
 - c. At 95% confidence, determine the critical value of F.
 - d. Using the critical value approach, test to determine whether there is a significant difference in the means of the three populations.
 - e. Determine the p -value and use it for the test.
2. Information regarding the starting salaries (in \$1,000) of samples of students in four different majors is given below.

	Majors			
	Management	Marketing	Finance	Accounting
Sample size	12	10	9	14
Average	26	31	22	25
Sample variance	10	7	15	9

- a. Compute the overall sample mean $\bar{X}$.
 - b. Set up the ANOVA table for this problem including the test statistic.
 - c. At 95% confidence, determine the critical value of F.
 - d. Using the critical value approach, test to determine whether there is a significant difference in the means of the three populations.
 - e. Determine the p -value and use it for the test.
3. Guitars R. US has three stores located in three different areas. Random samples of the sales of the three stores (in \$1000) are shown below.

Store 1	Store 2	Store 3
80	85	79
75	86	85
76	81	88
89	80	
80		

- a. Compute the overall mean $\bar{X}$.
- b. State the null and alternative hypotheses to be tested.
- c. Show the complete ANOVA table for this test including the test statistic.
- d. The null hypothesis is to be tested at 95% confidence. Determine the critical value for this test. What do you conclude?
- e. Determine the p -value and use it for the test.

4. The heating bills for a selected sample of houses using various forms of heating are given below. (Values are in dollars.)

Natural Gas	Central Electric	Heat Pump
84	95	85
64	60	93
93	89	90
88	96	92
71	90	80

- At $\alpha = 0.05$, test to see if there is a significant difference among the average heating bills of the homes. Use the p -value approach.
 - Test the above hypotheses using the critical value approach. Let $\alpha = .05$.
5. In a completely randomized experimental design, 18 experimental units were used for the first treatment, 10 experimental units for the second treatment, and 15 experimental units for the third treatment. Part of the ANOVA table for this experiment is shown below.

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F
Between Treatments	_____?	_____?	_____?	3.0
Error (Within Treatments)	_____?	_____?	6	
Total	_____?	_____?		

- Fill in **all** the blanks in the above ANOVA table.
 - At 95% confidence, test to see if there is a significant difference among the means.
6. Random samples were selected from three populations. The data obtained are shown below. ***Please note that the sample sizes are not equal.***

Treatment 1	Treatment 2	Treatment 3
37	43	28
33	39	32
36	35	33
38	38	
	40	

- Compute the overall mean $\bar{X}$.
 - At 95% confidence using the critical value and p -value approach, test to see if there is a significant difference among the means.
7. In a completely randomized experimental design, 7 experimental units were used for the first treatment, 9 experimental units for the second treatment, and 14 experimental units for the third treatment. Part of the ANOVA table for this experiment is shown below.

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F
Between Treatments	_____?	_____?	_____?	4.5
Error (Within Treatments)	_____?	_____?	4	
Total	_____?	_____?		

- Fill in **all** the blanks in the above ANOVA table.
- At 95% confidence using both the critical value and p -value approaches, test to see if there is a significant difference among the means.

8. Random samples were selected from three populations. The data obtained are shown below. ***Please note that the sample sizes are not equal.***

Treatment 1	Treatment 2	Treatment 3
45	31	39
41	34	35
37	35	40
40	40	
42		

- Compute the overall mean $\bar{X}$.
 - At 95% confidence, test to see if there is a significant difference among the means.
9. In order to compare the life expectancies of three different brands of printers, eight printers of each brand were randomly selected. Information regarding the three brands is shown below.

	Brand A	Brand B	Brand C
Average life (in months)	62	52	60
Sample variance	36	25	49

- Compute the overall mean $\bar{X}$.
 - State the null and alternative hypotheses to be tested.
 - Show the complete ANOVA table for this test including the test statistic.
 - The null hypothesis is to be tested at 95% confidence. Determine the critical value for this test. What do you conclude?
 - Determine the p -value and use it for the test.
10. Six observations were selected from each of three populations. The data obtained is shown below.

Sample 1	Sample 2	Sample 3
31	37	37
28	32	31
34	34	32
32	24	39
26	32	30
29	33	35

Test at the $\alpha = 0.05$ level to determine if there is a significant difference in the means of the three populations. Use both the critical value and the p -value approaches.

11. The test scores for selected samples of sociology students who took the course from three different instructors are shown below.

Instructor A	Instructor B	Instructor C
83	90	85
60	55	90
80	84	90
85	91	95
71	85	80

At $\alpha = 0.05$, test to see if there is a significant difference among the averages of the three groups. Use both the critical value and p -value approaches.

12. Three universities administer the same comprehensive examination to the recipients of MS degrees in psychology. From each institution, a random sample of MS recipients was selected, and these recipients were then given the exam. The following table shows the scores of the students from each university. Note that the sample sizes are not equal.

University A	University B	University C
89	60	82
95	95	70
75	89	90
92	80	79
99	66	
90		

- Compute the overall mean $\bar{X}$.
 - At $\alpha = 0.01$, test to see if there is any significant difference in the average scores of the students from the three universities. Use both the critical value and p -value approaches.
13. In a completely randomized experimental design, 11 experimental units were used for each of the 3 treatments. Part of the ANOVA table is shown below.

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Squares	F
Between Treatments	1,500	_____?	_____?	_____?
Within Treatments (Error)	_____?	_____?	_____?	
Total	6,000			

- Fill in the blanks in the above ANOVA table.
- At 95% confidence, test to determine whether or not the means of the 3 populations are equal.

14. Carolina, Inc. has three stores located in three different areas. Random samples of the sales of the three stores (in \$1,000) are shown below. Please note that the sample sizes **are not equal**.

Store 1	Store 2	Store 3
88	76	85
84	78	67
88	60	58
82	62	
93		

- a. Compute the overall mean $\bar{X}$.
- b. At 95% confidence, test to see if there is a significant difference in the average sales of the three stores. Use both the critical value and p -value approaches. Show your complete work and the ANOVA table.
15. In order to compare the life expectancies of three different brands of tires, ten tires of each brand were randomly selected and were subjected to standard wear testing procedures. Information regarding the three brands is shown below.

	Brand A	Brand B	Brand C
Average mileage (in 1000 miles)	37	38	33
Sample variance	3	4	2

Use the above data and test to see if the mean mileage for all three brands of tires is the same. Let $\alpha = 0.05$. Use both the critical value and p -value approaches.

16. Three different models of automobiles (A, B, and C) were compared for gasoline consumption. For each model of car, fifteen cars were randomly selected and subjected to standard driving procedures. The average miles/gallon obtained for each model of car and sample standard deviations are shown below.

	Car A	Car B	Car C
Average Mile/Gallon	42	49	44
Sample Standard Deviation	4	5	3

Use the above data and test to see if the mean gasoline consumption for all three models of cars is the same. Let $\alpha = 0.05$, and use both the critical and p -value approaches.

17. At $\alpha = 0.05$, test to determine if the means of the three populations (from which the following samples are selected) are equal. Use both the critical and p -value approaches.

Sample 1	Sample 2	Sample 3
60	84	60
78	78	57
72	93	69
66	81	66

18. In order to test to see if there is any significant difference in the mean number of units produced per week by each of three production methods, the following data were collected. (Please note that the sample sizes are not equal.)

Method I	Method II	Method III
182	170	162
170	192	166
179	190	

- Compute $\bar{X}$.
 - At the $\alpha = 0.05$ level of significance, is there any difference in the mean number of units produced per week by each method? Show the complete ANOVA table. Use both the critical and p -value approaches.
19. A dietician wants to see if there is any difference in the effectiveness of three diets. Eighteen people were randomly chosen for the test. Then each individual was randomly assigned to one of the three diets. Below you are given the total amount of weight lost in six months by each person.

Diet A	Diet B	Diet C
14	12	25
18	10	32
20	22	18
12	12	14
20	16	17
18	12	14

- State the null and alternative hypotheses.
 - Calculate the test statistic.
 - What would you advise the dietician about the effectiveness of the three diets? Use a .05 level of significance.
20. The Ahmadi Corporation wants to increase the productivity of its line workers. Four different programs have been suggested to help increase productivity. Twenty employees, making up a sample, have been randomly assigned to one of the four programs and their output for a day's work has been recorded. You are given the results below.

Program A	Program B	Program C	Program D
150	150	185	175
130	120	220	150
120	135	190	120
180	160	180	130
145	110	175	175

- State the null and alternative hypotheses.
 - Construct an ANOVA table.
 - As the statistical consultant to Ahmadi, what would you advise them? Use a .05 level of significance. Use both the critical and p -value approaches.
 - Use Fisher's LSD procedure and determine which population mean (if any) is different from the others. Let $\alpha = .05$.
21. The marketing department of a company has designed three different boxes for its product. It wants to determine which box will produce the largest amount of sales. Each box will be test marketed in five different stores for a period of a month. Below you are given the information on sales.

	Store 1	Store 2	Store 3	Store 4	Store 5
Box 1	210	230	190	180	190
Box 2	195	170	200	190	193
Box 3	295	275	290	275	265

- State the null and alternative hypotheses.
- Construct an ANOVA table.
- What conclusion do you draw?
- Use Fisher's LSD procedure and determine which mean (if any) is different from the others.
Let $\alpha = 0.01$.

22. You are given an ANOVA table below with some missing entries.

Source Variation	Sum of Squares	Degrees of Freedom	Mean Square	F
Between Treatments		3	1,198.8	
Between Blocks	5,040	6	840	
Error	5,994	18		
Total		27		

- State the null and alternative hypotheses.
- Compute the sum of squares between treatments.
- Compute the mean square due to error.
- Compute the total sum of squares.
- Compute the test statistic F.
- Test the null hypothesis stated in Part a at the 1% level of significance. Be sure to state your conclusion.

23. For four populations, the population variances are assumed to be equal. Random samples from each population provide the following data.

Population	Sample Size	Sample Mean	Sample Variance
1	11	40	23.4
2	11	35	21.6
3	11	39	25.2
4	11	37	24.6

Using a .05 level of significance, test to see if the means for all four populations are the same.

24. A research organization wishes to determine whether four brands of batteries for transistor radios perform equally well. Three batteries of each type were randomly selected and installed in the three test radios. The number of hours of use for each battery is given below.

Radio	Brand			
	1	2	3	4
A	25	27	20	28
B	29	38	24	37
C	21	28	16	19

- Use the analysis of variance procedure for completely randomized designs to determine whether there is a significant difference in the mean useful life of the four types of batteries.

(Ignore the fact that there are different test radii.) Use the .05 level of significance and be sure to construct the ANOVA table.

- b. Now consider the three different test radii and carry out the analysis of variance procedure for a randomized block design. Include the ANOVA table.
 - c. Compare the results in Parts a and b.
25. Employees of MNM Corporation are about to undergo a retraining program. Management is trying to determine which of three programs is the best. They believe that the effectiveness of the programs may be influenced by sex. A factorial experiment was designed. You are given the following information.

Factor A: Program	Factor B: Sex	
	Male	Female
Program A	320	380
	240	300
Program B	160	240
	180	210
Program C	240	360
	290	380

- a. Set up the ANOVA table.
 - b. What advice would you give MNM? Use a .05 level of significance.
26. The final examination grades of random samples of students from three different classes are shown below.

Class A	Class B	Class C
92	91	85
85	85	93
96	90	82
95	86	84

At the $\alpha = .05$ level of significance, is there any difference in the mean grades of the three classes?

27. Individuals were randomly assigned to three different production processes. The hourly units of production for the three processes are shown below.

Production Process		
Process 1	Process 2	Process 3
33	33	28
30	35	36
28	30	30
29	38	34

Use the analysis of variance procedure with $\alpha = 0.05$ to determine if there is a significant difference in the mean hourly units of production for the three types of production processes. Use both the critical and p -value approaches.

28. Random samples of employees from three different departments of MNM Corporation showed the following yearly incomes (in \$1,000).

Department A	Department B	Department C
40	46	46
37	41	40
43	43	41
41	33	48
35	41	39
38	42	44

At $\alpha = .05$, test to determine if there is a significant difference among the average incomes of the employees from the three departments. Use both the critical and p -value approaches.

29. The heating bills for a selected sample of houses using various forms of heating are given below (values are in dollars).

Gas Heated Homes	Central Electric	Heat Pump
83	90	81
80	88	83
82	87	80
83	82	82
82	83	79

At $\alpha = 0.05$, test to see if there is a significant difference among the average bills of the homes. Use both the critical and p -value approaches.

30. Three universities in your state decided to administer the same comprehensive examination to the recipients of MBA degrees from the three institutions. From each institution, MBA recipients were randomly selected and were given the test. The following table shows the scores of the students from each university.

Northern University	Central University	Southern University
75	85	80
80	89	81
84	86	84
85	88	79
81		83
		85

At $\alpha = 0.01$, test to see if there is any significant difference in the average scores of the students from the three universities. (Note that the sample sizes are not equal.) Use both the critical and p -value approaches.

31. The three major automobile manufacturers have entered their cars in the Indianapolis 500 race. The speeds of the tested cars are given below.

Manufacturer A	Manufacturer B	Manufacturer C
180	177	175
175	180	176
179	167	177

176
 190

172

At $\alpha = .05$, test to see if there is a significant difference in the average speeds of the cars of the auto manufacturers. Use both the critical and p -value approaches.

32. Part of an ANOVA table is shown below.

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F
Between Treatments	90	3	_____?	_____?
Within Treatments (Error)	120	20	_____?	

- Compute the missing values and fill in the blanks in the above table. Use $\alpha = .01$ to determine if there is any significant difference among the means.
- How many groups have there been in this problem?
- What has been the total number of observations?

33. Part of an ANOVA table involving 8 groups for a study is shown below.

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F
Between Treatments	126	_____?	_____?	_____?
Within Treatments (Error)	240	_____?	_____?	
Total	_____?	67		

- Complete all the missing values in the above table and fill in the blanks.
- Use $\alpha = 0.05$ to determine if there is any significant difference among the means of the eight groups.

34. MNM, Inc. has three stores located in three different areas. Random samples of the daily sales of the three stores (in \$1,000) are shown below.

Store 1	Store 2	Store 3
9	10	6
8	11	7
7	10	8
8	13	11

At 95% confidence, test to see if there is a significant difference in the average sales of the three stores. Use both the critical and p -value approaches.

35. Five drivers were selected to test drive 2 makes of automobiles. The following table shows the number of miles per gallon for each driver driving each car.

Automobile	Drivers				
	1	2	3	4	5
A	30	31	30	27	32
B	36	35	28	31	30

Consider the makes of automobiles as treatments and the drivers as blocks, test to see if there is any difference in the miles/gallon of the two makes of automobiles. Let $\alpha = .05$.

36. A factorial experiment involving 2 levels of factor A and 2 levels of factor B resulted in the following.

Factor A	Factor B	
	Level 1	Level 2
	14	18
	16	12
Level 2	18	16
	20	14

Set up an ANOVA table and test for any significant main effect and any interaction effect. Use $\alpha = .05$.

37. Ten observations were selected from each of 3 populations, and an analysis of variance was performed on the data. The following are the results:

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F
Between Treatments	82.4			
Within Treatments (Error)	158.4			

- Using $\alpha = .05$, test to see if there is a significant difference among the means of the three populations.
 - If in Part a you concluded that at least one mean is different from the others, determine which mean is different. The three sample means are $\bar{x}_1 = 24.8$, $\bar{x}_2 = 23.4$, and $\bar{x}_3 = 27.4$.
38. The following are the results from a completely randomized design consisting of 3 treatments.

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F
Between Treatments	390.58			
Within Treatments (Error)	158.4			
Total	548.98	23		

- Using $\alpha = .05$, test to see if there is a significant difference among the means of the three populations. The sample sizes for the three treatments are equal.
- If in Part a you concluded that at least one mean is different from the others, determine which mean(s) is(are) different. The three sample means are $\bar{x}_1 = 17.000$, $\bar{x}_2 = 21.625$, and $\bar{x}_3 = 26.875$. Use Fisher's LSD procedure and let $\alpha = .05$.

39. Eight observations were selected from each of 3 populations (total of 24 observations), and an analysis of variance was performed on the data. The following are part of the results.

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F
Between Treatments	216			
Within Treatments (Error)	252			

Using $\alpha = .05$, test to see if there is a significant difference among the means of the three populations.

40. Random samples of individuals from three different cities were asked how much time they spend per day watching television. The results (in minutes) for the three groups are shown below.

City I	City II	City III
260	178	211
280	190	190
240	220	250
260	240	
300		

At $\alpha = 0.05$, test to see if there is a significant difference in the averages of the three groups.

41. Three different brands of tires were compared for wear characteristics. From each brand of tire, ten tires were randomly selected and subjected to standard wear-testing procedures. The average mileage obtained for each brand of tire and sample variances (both in 1,000 miles) are shown below.

	Brand A	Brand B	Brand C
Average Mileage	37	38	33
Sample Variance	3	4	2

At 95% confidence, test to see if there is a significant difference in the average mileage of the three brands.

42. Nancy, Inc. has three stores located in three different areas. Random samples of the sales of the three stores (In \$1,000) are shown below.

Store 1	Store 2	Store 3
46	34	33
47	36	31
45	35	35
42	39	
45		

- Compute the overall mean $\bar{X}$.
 - At 95% confidence, test to see if there is a significant difference in the average sales of the three stores.
43. In a completely randomized experimental design, 11 experimental units were used for each of the 4 treatments. Part of the ANOVA table is shown below.

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F
Between Treatments	1500	_____?	_____?	_____?
Within Treatments	_____?	_____?	_____?	
Total	5500			

Fill in the blanks in the above ANOVA table.

44. Samples were selected from three populations. The data obtained are shown below.

	Sample 1	Sample 2	Sample 3
	10	16	15
	13	14	18
	12	15	
	13		
Sample Mean ($\bar{x}_j$)	12	15	16.5
Sample Variance (s_j^2)	2.0	1.0	4.5

- Compute the overall mean $\bar{X}$.
 - Set up an ANOVA table for this problem.
 - At 95% confidence, test to determine whether there is a significant difference in the means of the three populations. Use both the critical and p -value approaches.
45. In a completely randomized experimental design, 14 experimental units were used for each of the 5 levels of the factor (i.e., 5 treatments). Fill in the blanks in the following ANOVA table.

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F
Between Treatments	_____?	_____?	800?	_____
Error (Within Treatments)	_____?	_____?	_____	
Total	10600?	_____?		

46. Random samples of several days' sales of handguns per day in three different states are shown below. We are interested in determining whether or not there is a significant difference in the average sales of guns in the three states

Tennessee	Kentucky	Texas
12	15	16
13	19	18

17 20
10
18

- a. Compute the overall mean $\bar{X}$.
 - b. State the null and alternative hypotheses to be tested.
 - c. Show the complete ANOVA table for this test including the test statistic.
 - d. The null hypothesis is to be tested at 95% confidence. Determine the critical value for this test. What do you conclude?
 - e. Determine the p -value and use it for the test.
47. Information regarding random samples of annual salaries (in thousands of dollars) of doctors in three different specialties is shown below.

	Pediatrics	Radiology	Pathology
Sample size	12	10	11
Average salary	120	186	240
Sample variance	16	18	20

- a. Compute the overall mean $\bar{X}$.
- b. State the null and alternative hypotheses to be tested.
- c. Show the complete ANOVA table for this test including the test statistic.
- d. The null hypothesis is to be tested at 95% confidence. Determine the critical value for this test. What do you conclude?
- e. Determine the p -value and use it for the test.